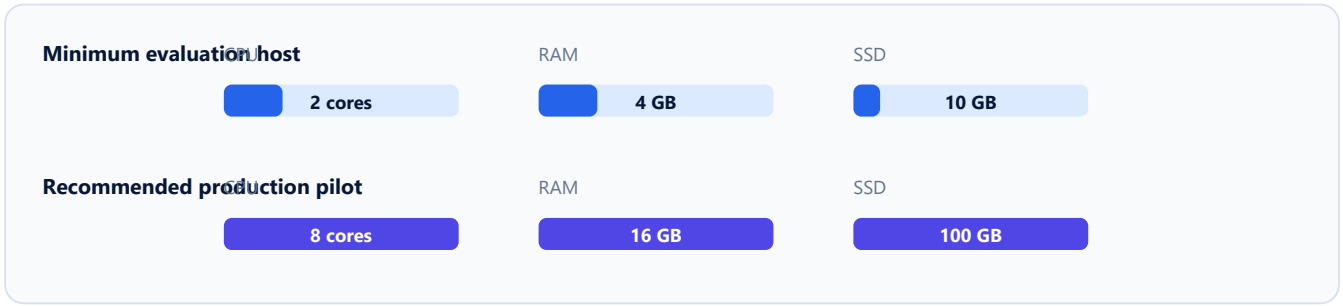
SENTRO

DSPM System Requirements

Minimum and recommended deployment sizing

Decision in one line

Use 2 CPU cores / 4 GB RAM / 10 GB free SSD only for a local evaluation. For a production pilot, start at 8 vCPU / 16 GB RAM / 100 GB fast SSD, use PostgreSQL, and validate against the real corpus and retention policy.



Assessment date	**10 September 2026**
Repository snapshot	**sentro-ironhack @ 5a2d5a0, including current working-tree changes**
Product stage	**Prototype / MVP for demos, local pilots, and product development**
Basis	**Static architecture review + repository benchmarks + local runtime/storage observation**

This report distinguishes a functional minimum from a recommended operating baseline. The numbers are engineering recommendations derived from the present codebase; they are not a formal vendor hardware certification.

SECTION 01

Executive requirements matrix

The minimum column is intended for one administrator evaluating local or small pilot scans. It is not a production profile. The recommended column is the safe starting point for a production pilot with compliance classification, scheduled operational work, and modest concurrent use.

Area	Minimum — local evaluation	Recommended — production pilot
CPU	2 x64 cores	8 vCPU; reserve capacity for scanner + detector processes
Memory	4 GB RAM 8 GB when OCR or ML is enabled	16 GB RAM 32 GB if OCR + ML + DB/search are colocated
Storage	10 GB free SSD; persistent local `data/`	100 GB initial NVMe/SSD, 30% free headroom; backups on separate durable storage
Host OS	64-bit Windows or Linux for core web/SMB use	64-bit Windows Server for the complete Windows endpoint workflow; Windows Server 2019 is repository-observed
Python	CPython 3.10–3.13, 64-bit	CPython 3.13, 64-bit, pinned runtime dependencies
Database	Built-in SQLite; one application worker	PostgreSQL; repository certification profile names 18.4; pooled connections
Application workers	1 Uvicorn worker	Start with 1 for scan-heavy use; 2–4 only for read-heavy service on PostgreSQL after load test
Browser	Current desktop Edge, Chrome, or Firefox; JavaScript and cookies enabled	Current managed desktop browser, 8 GB client RAM, 1920×1080 display
Network	TCP 8000 locally; TCP 445 for SMB; 5985/5986 for WinRM as used	Low-latency LAN to targets; TLS/reverse proxy; TCP 5432 to PostgreSQL and 7700 to optional Meilisearch
Intended scale	Demo / small pilot; one active scan	Production pilot; scale and retention verified with the customer corpus

Production database boundary

In the current implementation, DSPM_ENV=production refuses SQLite. PostgreSQL is therefore a functional production prerequisite, not merely a performance preference. SQLite remains appropriate for local evaluation and development.

Interpretation notes

- A ‘core’ means a schedulable x86-64 CPU core or vCPU. Avoid heavily oversubscribed burst instances for compliance scans.
- SSD capacity means free, durable space after OS and update reserve. The scanner’s database, caches, report artifacts, and backups grow independently.
- If the application, PostgreSQL, and Meilisearch share one host, add their CPU, memory, and storage needs; do not reuse the app-only minimum.

SECTION 02

Product and workload analysis

Sentro is a multi-tenant Data Security Posture Management (DSPM) and DLP-readiness console. It inventories local paths, SMB file servers, and Windows endpoints over WinRM; extracts bounded content; classifies sensitive data and compliance signals; scores risk; stores normalized inventory and scan history; and exports evidence and remediation reports.

Component	Implementation	Sizing behavior
Web/API	FastAPI + Uvicorn; static HTML/CSS/JavaScript console	Light at idle; concurrency increases per-process caches and DB connections
Discovery	Local filesystem, SMB, and WinRM collectors	I/O- and per-file-metadata bound; latency and ACL calls can dominate
Classification	Regex/heuristic + GDPR, PCI DSS, NIST; optional spaCy ML	CPU-bound; process pool uses all but one core by default on hosts with ≥3 cores
OCR	Optional RapidOCR/ONNX Runtime; local and SMB images	~150 MB install; approximately 2 s/image on CPU; serialized inference by default
Persistence	SQLite local mode; PostgreSQL production mode; encrypted report artifact tier	Retention dominates disk; multiple workers favor PostgreSQL
Search	Database search by default; optional Meilisearch v1.52.0 projection	Separate index volume and memory budget at large inventory scale
Client	Desktop browser; server-side paginated inventory and streaming exports	Bounded pages protect browser memory; large report rendering still benefits from a modern client

What drives each resource

CPU
Compliance detectors are pure-Python CPU work. The product uses a process pool so additional cores materially improve scan time on large, compliance-heavy corpora.

Memory
Base service memory is modest, but OCR/ML runtimes, multiple Uvicorn workers, PostgreSQL caches, and one copy of the large compliance catalog per detector process add up.

Disk
The system stores inventory rows, encrypted report bodies, caches, generated reports, backups, and optional search indexes. Retention policy is more important than source-code size.

Network
SMB and WinRM performance depends on round-trip time, target metadata calls, and available bandwidth. The repository's modeled 100 ms WinRM case gained 3.37× from six workers.

Platform implication
The core API can run on Windows or Linux, but local endpoint scanning is explicitly Windows-only. Use a Windows Server host when full endpoint functionality and the supplied Windows/PostgreSQL administration scripts are required.

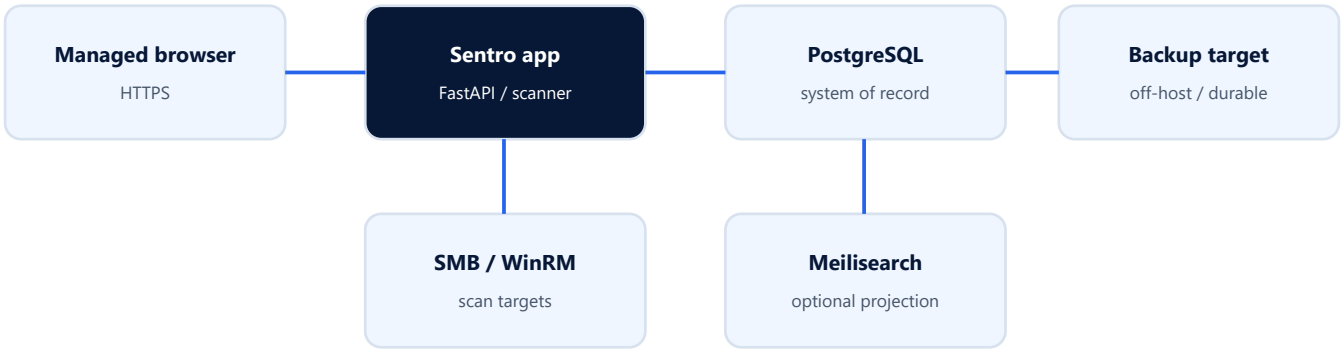
SECTION 03

Deployment profiles

These profiles turn the minimum/recommended numbers into deployable configurations. They are starting points; the acceptance test in Section 07 determines whether a specific estate is ready.

Profile	App / scanner	Data services	Use boundary
A — Evaluation	2 cores 4 GB RAM 10 GB SSD 1 worker	SQLite on local SSD	Demo, development, small pilot, one active scan; base classifiers
B — Production pilot	8 vCPU 16 GB RAM 100 GB initial SSD 1–2 workers	PostgreSQL on durable SSD; separate backups	Modest multi-user use, compliance scans, operational history; load-test first
B+ — Feature-heavy	8–12 vCPU 32 GB RAM 150+ GB SSD	PostgreSQL; optional Meilisearch on separate service/volume	OCR + ML, longer retention, parallel scans, or colocated services
C — Large inventory	Separate app/scanner tier 8–16 vCPU 16–32 GB	Separate PostgreSQL: 8–16 vCPU, 32–64 GB; NVMe; optional separate Meilisearch	Million-row estates. Capacity and concurrency must be re-certified on production-like hardware

Recommended topology



Worker guidance

- Use one Uvicorn worker for a scan-heavy pilot unless concurrent browser/API demand proves it insufficient.
- Multiple Uvicorn workers disable several in-process caches for correctness and can multiply memory use. Use PostgreSQL and measure before increasing to 2–4.
- The detector process pool defaults to roughly CPU minus one and disables itself below three cores. Multiple simultaneous scans can oversubscribe CPU; cap detector processes when concurrency is enabled.

SECTION 04

Evidence behind the numbers

The recommendation is anchored in measurements already present in the repository plus a local observation of this checkout. It deliberately leaves enough headroom for the operating system, child processes, scanning bursts, and database/cache growth.

155.6 MB	30.1 s	5.0 M	5,377/s
Observed idle app working set*	2,160-file full compliance scan	Synthetic inventory rows certified	8-worker DB soak operations

Evidence	Observed result	Sizing implication
Local runtime probe*	One app process, SQLite (171 MB), OCR dependency enabled: 155.6 MB working set; health response 80 ms	Base service fits well below 4 GB, but this is idle—not peak scan memory
Cold local scan	424 tiny files: 168.7 s, 2.5 files/s, 33.25 MB peak traced Python memory	Per-file metadata/ACL overhead matters more than data volume for tiny files
Six-core compliance run	2,160 files, GDPR + PCI DSS + NIST: 30.1 s after process parallelism vs 412.4 s original	Extra cores are valuable; an 8-vCPU production start leaves room for parent + OS
Modeled WinRM latency	80 files × 100 ms: 13.76 s at 1 worker, 4.08 s at 6 workers; 3.37× speedup	Network latency and target behavior must be represented in acceptance testing
PostgreSQL certification	5M rows; 8 readers for 300 s; 1,613,268 ops; zero errors; 2.69 ms p95 / 3.67 ms p99	Architecture scales, but the artifact omits hardware telemetry and cannot certify a hardware minimum
Repository storage observation*	171 MB DB + 450 MB backups + 339 MB cache + 318 MB reports + 55 MB artifacts = ~1.30 GB	Operational derivatives exceed the live DB; 10 GB is only a small-pilot floor

* Local observation on the analysis host: Windows Server 2019 x64, Intel Xeon Silver 4215R, 6 cores, 16 GB RAM. The installed interpreter was Python 3.14.6, which is outside the declared product range; the probe demonstrates observed behavior only and does not expand support beyond Python 3.10–3.13.

Why 16 GB recommended when idle memory is ~156 MB

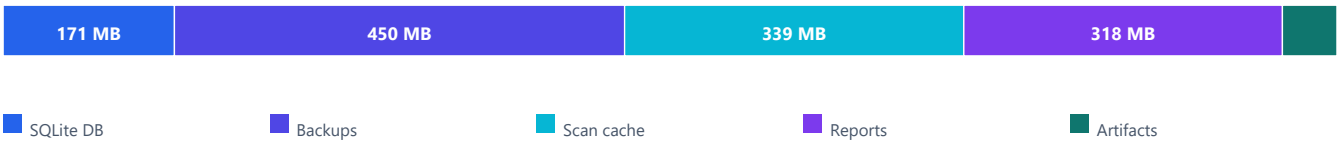
An active compliance scan can spawn multiple detector processes, each with its own catalog; OCR/ML add native runtimes; PostgreSQL and optional search need caches; and multiple application workers duplicate process state. The recommendation budgets for those peaks and preserves operating-system file cache instead of sizing to idle RSS.

SECTION 05

Storage and capacity planning

Do not size Sentro from repository size or database size alone. The durable footprint includes several independently retained data sets.

Observed workspace operational data: 1.30 GB



Observed values are from this working checkout on 10 September 2026 and include development activity. They illustrate composition, not a universal production multiplier.

Capacity formula

Usable data volume ≥ database + encrypted artifacts + scan cache + generated reports + retained backups + search index + 30% free headroom

Planning item	Minimum treatment	Recommended treatment
Primary data	Local SSD, persistent `data/` path	Dedicated encrypted SSD/NVMe volume with monitoring
Backups	At least one recoverable copy outside active data directory	Encrypted, off-host copies; retention and restore drills sized separately
Free space	Never consume the final 10 GB allocation	Alert before 70% used; retain ≥30% free for indexes, WAL, exports, and maintenance
SQLite	Local filesystem only	Not a production option in current code; do not place WAL database on SMB/NFS
PostgreSQL	N/A for evaluation	Provision DB + indexes + WAL + backup staging; use measured customer ingestion
Search	Database search	If Meilisearch is enabled, allocate an independent persistent index/snapshot volume

Sizing procedure

- Run a representative scan containing the customer's real file-size and extension distribution, with the intended compliance frameworks and OCR/ML settings.
- Measure the database, artifact, cache, report, and backup directories after at least two full scans plus one unchanged incremental scan.
- Multiply by the intended tenant count and retained scan generations, then add the optional search index and 30% free-space headroom.
- Repeat after retention and legal-hold policy is finalized; legal holds can deliberately prevent deletion.

SECTION 06

Software, network, and client prerequisites

Required and optional software

Layer	Minimum	Recommended / feature condition
Runtime	64-bit CPython 3.10–3.13; pip; pinned `requirements.txt`	Python 3.13 in an isolated virtual environment; hash-verified lock file where deployment policy requires
Database	Python built-in SQLite for local mode	PostgreSQL; profile currently approves major 18, minimum 18.4; psycopg 3.3.4 + pool 3.3.1
Console	No Node.js build required; static assets are served by FastAPI	Current managed Edge/Chrome/Firefox; browser printing with background graphics for best PDF output
ML / NER	Off by default	spaCy 3.8.7 + `en_core_web_sm`; enabling changes detection output
OCR	Off by default in repository configuration	RapidOCR 1.2.3 / ONNX Runtime; use 8+ GB RAM at minimum and benchmark image mix
Search	Database provider	Meilisearch v1.52.0 deployment is supplied; container runtime required for the compose workflow
Windows endpoint	pywinrm 0.5.0; reachable WinRM	Windows host for local endpoint scan; PowerShell/WinRM configured through GPO; Kerberos extra where enabled

Network paths

Port / protocol	Direction	Purpose	Condition
TCP 8000	Browser → Sentro	Default Uvicorn HTTP listener	Usually terminate TLS at a reverse proxy in production
TCP 445	Sentro → file server	SMB share discovery and content/metadata access	Required for SMB scans
TCP 5985	Sentro → endpoint	WinRM HTTP	Lab / policy-dependent
TCP 5986	Sentro → endpoint	WinRM HTTPS	Recommended and normally required in production
TCP 5432	Sentro → PostgreSQL	Production system of record	Required when DB is remote; TLS for non-local connections
TCP 7700	Sentro → Meilisearch	Optional inventory search projection	Bind privately; API key required

Target permissions are a functional prerequisite

Normal SMB shares require read access. Hidden administrative shares such as C$ or D$ require a suitable local administrator. Windows endpoint scans require a credential accepted by the target and reachable WinRM; WMI/DCOM may also be needed for one-time activation.

SECTION 07

Acceptance test and limitations

A deployment should be called ‘ready’ only after it meets these gates on production-like hardware, network paths, content, and retention settings.

Gate	Suggested pass condition	Why it matters
Functional	Local/SMB/WinRM scan paths in scope complete; findings and exports are correct	Hardware speed cannot compensate for unreachable targets or unsupported workflow
Scan throughput	Representative full and incremental scans finish within the customer’s maintenance window	Per-file ACL/metadata cost and compliance mix vary widely
Host headroom	Peak CPU <85%; memory has ≥25% available; no sustained paging	Detector processes and OCR create bursty peaks
Target impact	Target CPU, disk latency, network, and foreground p95 stay within customer budget	Repository explicitly lacks real NAS/WAN endpoint-impact certification
API / UI	Critical interactions and inventory searches meet agreed p95 under concurrent scan load	Read-heavy and scan-heavy workloads compete differently
Storage	Projected retention fits with ≥30% free space; backup and restore drill pass	Reports, caches, artifacts, WAL, and backups all grow
Recovery	Database plus encrypted artifact tier restore together; keys remain available separately	A database-only restore can leave report references without bodies

Known limits of this assessment

- The repository identifies itself as a prototype/MVP and says it should be hardened before production deployment.
- No formal OS/browser support matrix, concurrent-user SLA, peak scan-memory result, or production storage-per-asset coefficient is published.
- The 5M-row certification used a synthetic projected table and records query/soak results, but not the host CPU, RAM, disk model, or database footprint.
- The endpoint performance baseline explicitly does not cover real 1M/10M-file NAS/WAN estates, target resource impact, backup/AV coexistence, or 24–72 hour soak.
- Current working-tree changes were included. Re-run this assessment after material dependency, database, OCR/ML, or scanning-concurrency changes.

Final recommendation

Approve Profile B as the procurement/deployment baseline for a production pilot: 8 vCPU, 16 GB RAM, 100 GB fast persistent storage, PostgreSQL, durable backups, and a Windows Server host when full endpoint features are required. Treat Profile A only as an evaluation floor. Require a production-like acceptance run before committing an enterprise capacity or SLA.

SECTION 08

Method and source register

This is a repository-grounded engineering assessment. No external product claims were used. Requirements marked as 'required' come from code/configuration behavior; resource numbers are recommendations inferred from the evidence and stated operating boundaries.

ID	Repository source	What it supports
S1	README.md — Quick Start, Configuration, Dependencies, Windows Endpoint Scans, Status	Python range; SQLite/PostgreSQL modes; ports; optional features; MVP status
S2	requirements.txt; requirements-ml.txt; requirements-ocr.txt; requirements-postgres.txt	Pinned runtime, ML, OCR, and PostgreSQL dependencies; OCR footprint/cost note
S3	.env.example	Resource-affecting defaults and bounds: worker pools, body/OCR limits, DB pool, scan deadline, cache settings
S4	main.py; backend/app.py	Uvicorn worker behavior; production/SQLite refusal; Windows-only local endpoint scan
S5	docs/perf_baseline.md	Cold scan, memory, parallelism, WinRM latency model, endpoint-certification limitations
S6	docs/PERFORMANCE_AUDIT_AND_ROADMAP_2026-09.md	Six-core compliance results, worker/catalog behavior, database/UI characteristics
S7	docs/database/P7_CERTIFICATION_PROFILE.json	PostgreSQL 18.4 baseline; 5M rows; 8 readers; 300 s; 300 ms p95 target
S8	docs/database/evidence/capacity_soak_20260903T061000Z.json	5M-row latency, concurrency, soak throughput, zero-error result
S9	deploy/meilisearch/compose.yaml	Optional Meilisearch v1.52.0 service, port, persistence, snapshots
S10	Local read-only observation, 10 Sep 2026	Host facts, idle app working set/health timing, and workspace storage composition

Definitions

Minimum	Lowest practical resources for the limited evaluation scenario stated in this report; not an SLA or production approval.
Recommended	A conservative initial production-pilot baseline with headroom; still requires workload validation.
Production	DSPM_ENV is set to production, triggering fail-closed controls and the PostgreSQL requirement.
Inventory row	Metadata projection used for browsing/search; not equivalent to one complete stored file or one finding.

Report v1.0 • Generated from the shared workspace • Output: docs/Sentro_DSPM_System_Requirements_Report.pdf